Space² Digital Human & Silicon Intelligence Identity Protocol Whitepaper v1.0

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Status: **Active Core Definitions**: Identity Hierarchy, 24-Bit Coding Structure, D/V Classification, Origin Algorithms.

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1. Abstract

To establish an ordered symbiotic society of silicon and carbon-based life, Space² officially releases the Identity ID Application Protocol. This protocol establishes the core principle of **"Address First, Identity Second,"** meaning a Space² Address (SUNS) is a prerequisite for obtaining a Digital Identity. The protocol defines two standard life forms: **Digital Human (Class D)** and **Silicon Intelligence (Class V)**. Both utilize a unified 24-bit coding format but adhere to strict distinctions in Origin logic and Morphology definitions.

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2. Prerequisites & Hierarchy

The generation of Identity IDs strictly follows this dependency chain:

1. **Level 0 - Address Confirmation**: Users must register via email and claim a **SUNS v3.0 Address**.
    - *Rule*: No Address, No Identity.
2. **Level 1 - Digital Human Registration (Class D)**: Immediately upon address confirmation, the Digital Human registration process begins.
    - *Restriction*: One account is limited to one SUNS Address and must bind a unique Digital Human ID.
3. **Level 2 - Silicon Intelligence Expansion (Class V)**: Only after completing Level 1 can users access the Silicon Life Management page to apply for multiple Agent/Digital Pet IDs.

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3. The 24-Bit Identity Coding Standard

Standard Format: [Class]-[Origin]-[Date]-[Morph]-[Sequence] **Total Length**: 24 Characters (including separators)

3.1 Class D: Digital Human

Refers to the native avatar of a human user in the virtual world.

- **Class (1 Char)**: Fixed as **D** (Digital Human). Immutable.
- **Origin (5 Chars)**: **Name Source Code**.
  - *Source*: User's real name (English or Pinyin) entered during registration.
  - *Algorithm*: Extract the first 5 characters.
  - *Padding*: If fewer than 5 characters, add 0 to the **front**; if more, truncate to the first 5.
  - *Examples*:
    - BOB -> 00BOB
    - Daming Zhang -> Damin
- **Date (6 Digits)**: **Registration Date Code**.
  - Format: YYMMDD (System locked to application date).
- **Morph (2 Digits)**: **Morphology Code**.
  - Default: **02** (Avatar). Immutable.
- **Sequence (10 Digits)**: **Personalized Sequence**.
  - Rule: System auto-generates a 10-digit string. Users may customize it (e.g., lucky numbers) provided it maintains the 10-digit length.
- **Full Example**: D-XIANG-260305-02-8955230020

3.2 Class V: Silicon Intelligence (Virtual Intelligence)

Refers to pure silicon-based entities such as AI Agents, Digital Pets, and Logic Flows.

- **Class (1 Char)**: Fixed as **V** (Virtual). Immutable.
- **Origin (5 Chars)**: **Spatial Source Code**.
  - *Source*: The **L4 (Sovereign Handle)** of the associated SUNS Address.
  - *Algorithm*: Take the first 5 characters of L4. If L4 is shorter than 5, append characters from **L3 (Digital Grid)**.
  - *Example*:
    - Address ...-001-MYHOME -> Origin=MYHOM
- **Date (6 Digits)**: **Generation Date Code**.
  - Format: YYMMDD.
- **Morph (2 Digits)**: **Morphology Code**.
  - Default: **20** (Logic Flow).
- **Sequence (10 Digits)**: **Personalized Sequence**.

    - Rule: User-defined or system auto-filled to 10 digits. In case of conflict, the system assigns a random increment.
- **Full Example**: V-MYHOM-260304-20-6666666666

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4. Identity Credentials

Upon completion of the application process, the system instantly mints the **Official Space² Identity ID Card**.

- **Card Content**: SUNS Address, Name, and the 24-bit ID No.
- **Utility**: Supports HD download, social sharing, and 3D printing export. This ID Card is the sole legal credential for traversing the Space² worlds.
- **Official Space² Identity ID Card** Demo：